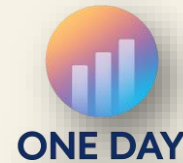


JOURNEY INTO AI-POWERED POWER BI DEVELOPMENT: INTEGRATING VS CODE, GITHUB COPILOT, AND CLAUDE CODE

How to use AI Coding Agents effectively



INTRODUCTION



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- ✓ 9+ Years of Power BI Experience
- ✓ 7+ Year of Training Experience
- ✓ Microsoft Certified Solution Expert – Data Management and Analytics
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- ✓ Fasttrack Recognized Solution Architect

POWER BI REPORT DEVELOPMENT BY CODE





```
Report/  
  definition/  
    pages/  
      Page1/  
        visuals/  
          Visual1/  
            visual.json  
            mobile.json  
          Visual2/  
            visual.json  
        page.json  
      pages.json  
    bookmarks/  
      bookmark1.bookmark.json  
      bookmarks.json  
    version.json  
    reportExtensions.json  
    report.json
```

OVERVIEW OF PBIR

Modular and Git-Friendly

PBIR breaks down reports into individual JSON files for pages, visuals, datasets, and more—making it ideal for source control systems like Git. This enables easier collaboration, version tracking, and conflict resolution.

Improved Developer Experience

With a documented schema and folder-based structure, PBIR allows developers to edit and manage reports using standard code editors like Visual Studio Code, streamlining development workflows.

Enhanced Reusability and Automation

Components such as visuals, bookmarks, and themes can be reused or programmatically generated, supporting automation and templating in enterprise-scale BI projects.



COMPLEXITIES OF THE PBIR FORMAT

Report folder structure

Components are separated into 100's of different files and folders.

Complex interactions between files

Specific settings that are easy to manage in the UI, could be extremely hard to code accurately.

Lack of detailed literature yet

Because of its novelty there is currently limited available resources that would help you learn how to effectively develop.



ROLE AND BENEFITS OF AI AGENTS IN DEVELOPMENT

Accelerated Code Generation

AI agents speed up code writing by automating repetitive programming tasks efficiently.

Enhanced Developer Creativity

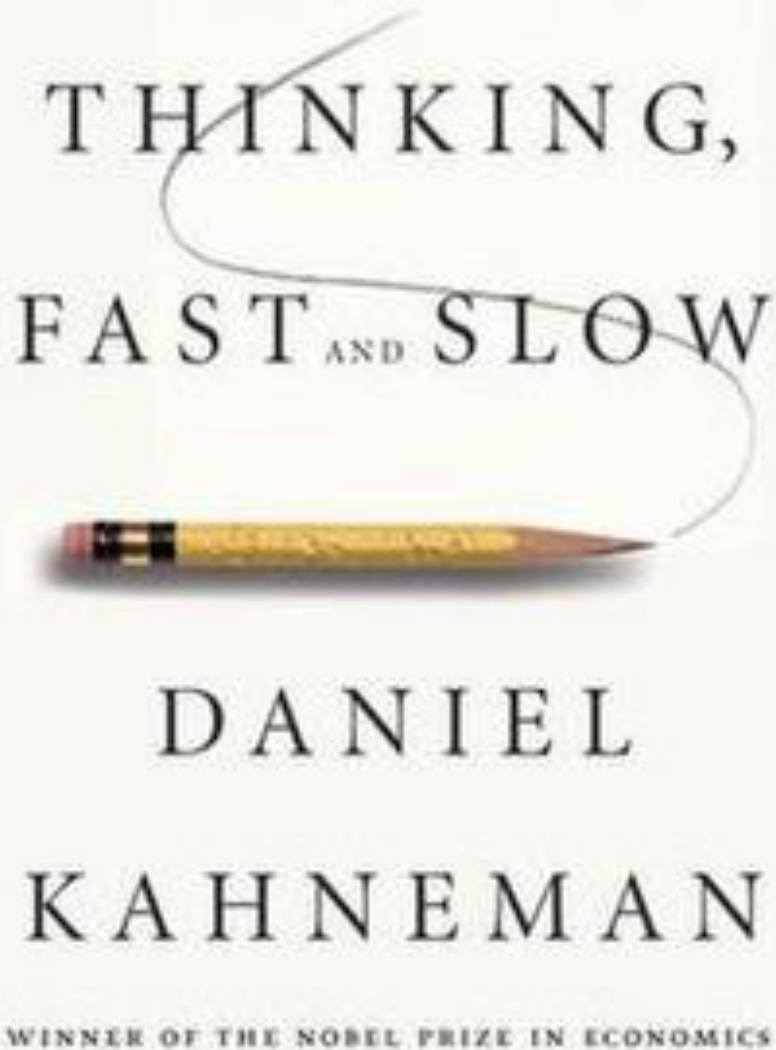
By handling routine tasks, AI allows developers to focus on innovative and strategic solutions.

Document first approach

AI tools require context, which force you to document in detail what you want to implement

HOW AI “THINKS”





SYSTEM 1 VS SYSTEM 2

Daniel Kahneman described in his book "Thinking, Fast and Slow":

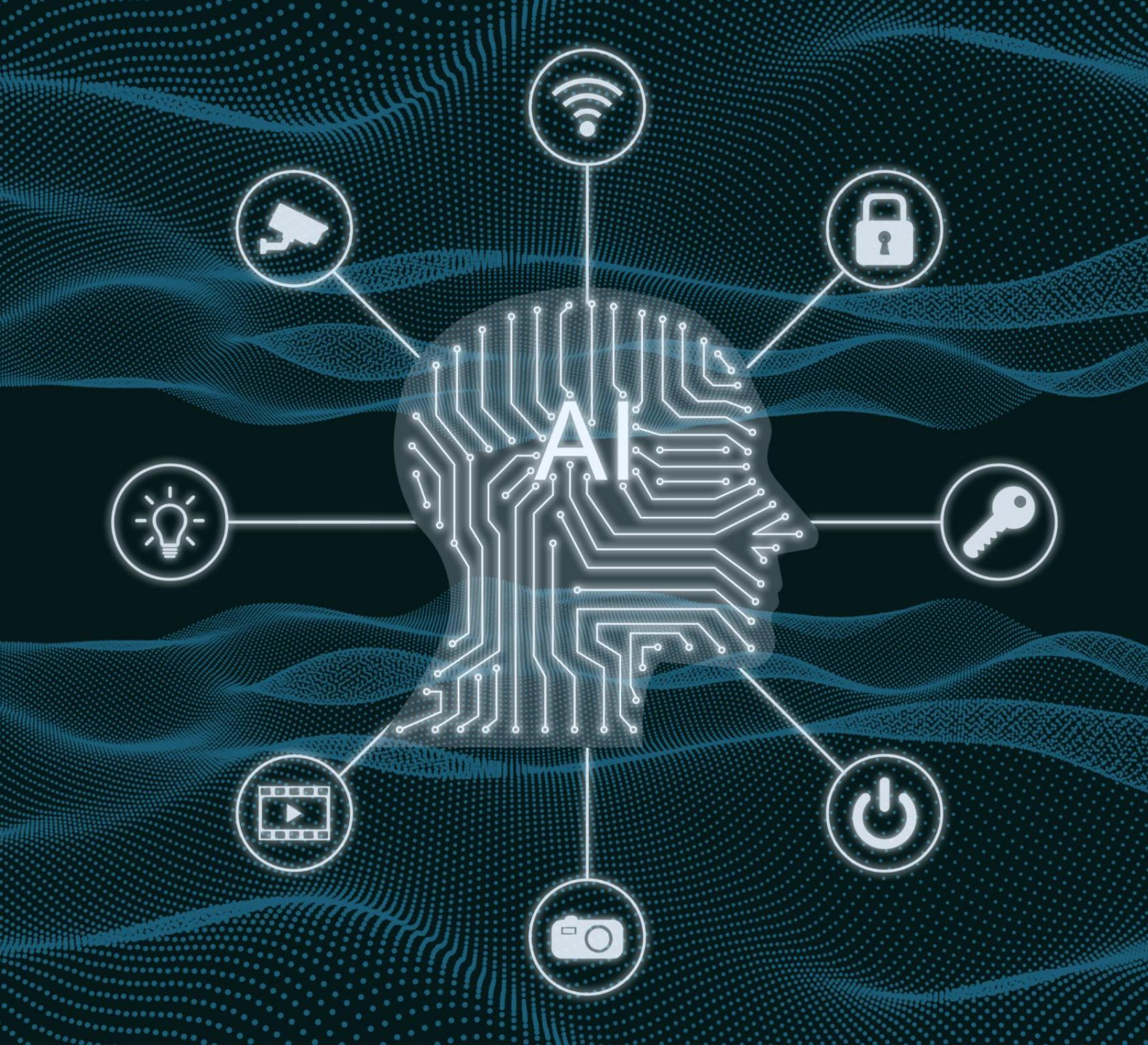
- **"System 1"** that is fast, instinctive and emotional
- **"System 2"** that is slower, more deliberative, and more logical

Walking, driving, shopping, generally going through life on “auto-pilot”, that is **System 1**.

But if you are learning something new or solve a non-trivial problem, it is **System 2**. When you are using **System 2** you feel it. It becomes hard and tiring.

Opinion:

LLMs are not thinking machines, but pattern matching algorithms. The reason we feel like they are thinking is because most of the time our brain is also mainly doing pattern matching.



WHAT DOES IT MEAN FOR A CODING AGENT

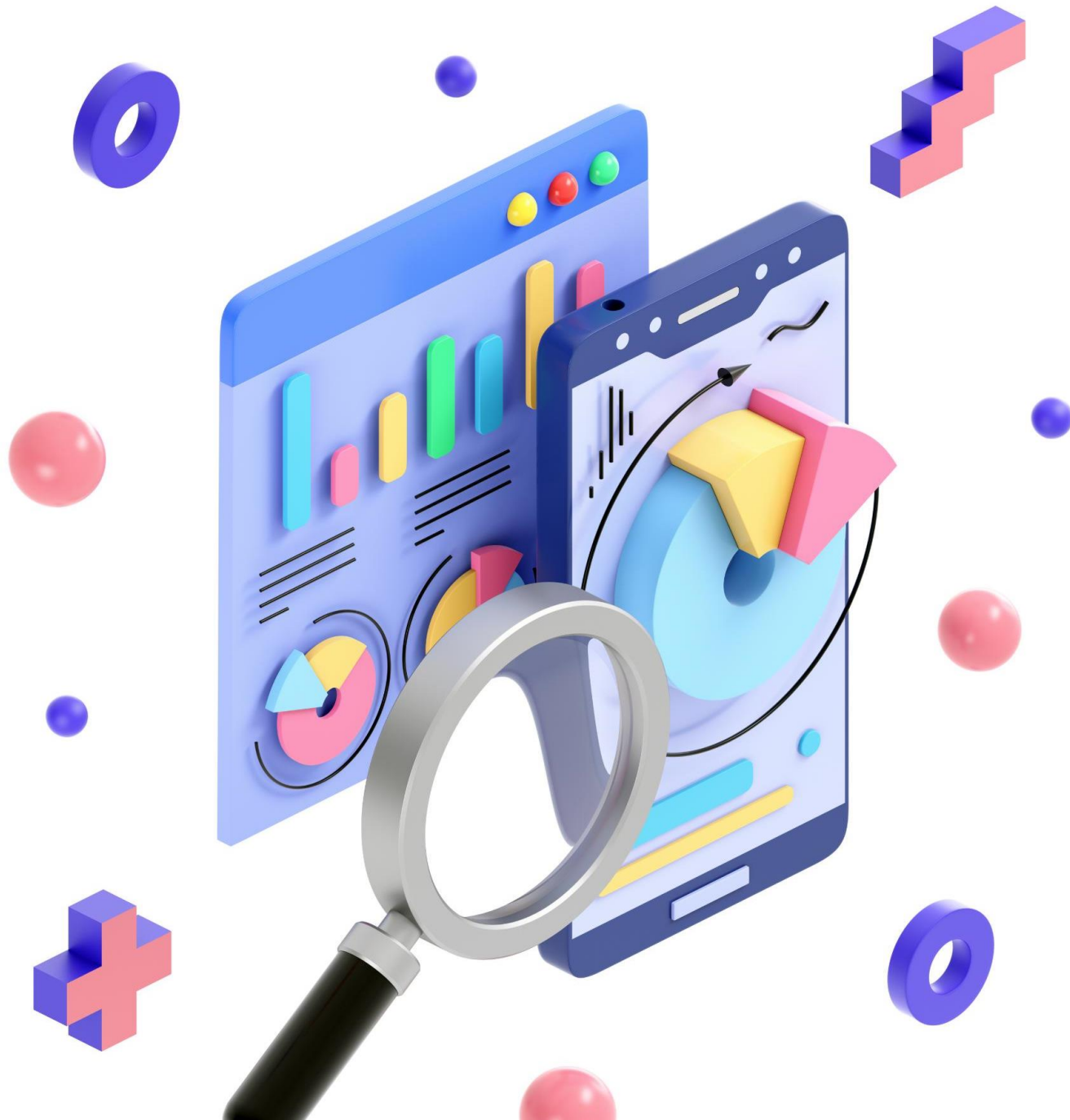
If the challenge you would like to solve does not have a huge literature or you want to solve problems in a novel way, then that is the time to use "System 2". But AI does not have a "System 2" option that works well.

The relatively newer reasoning models are one attempt to try to solve this problem, but they still require ample data in their training set to not start hallucinating.

This is the time for you to provide the scaffolding!

CONTEXT ENGINEERING





EVOLUTION OF INTERACTING WITH AI AGENTS

Prompt Engineering

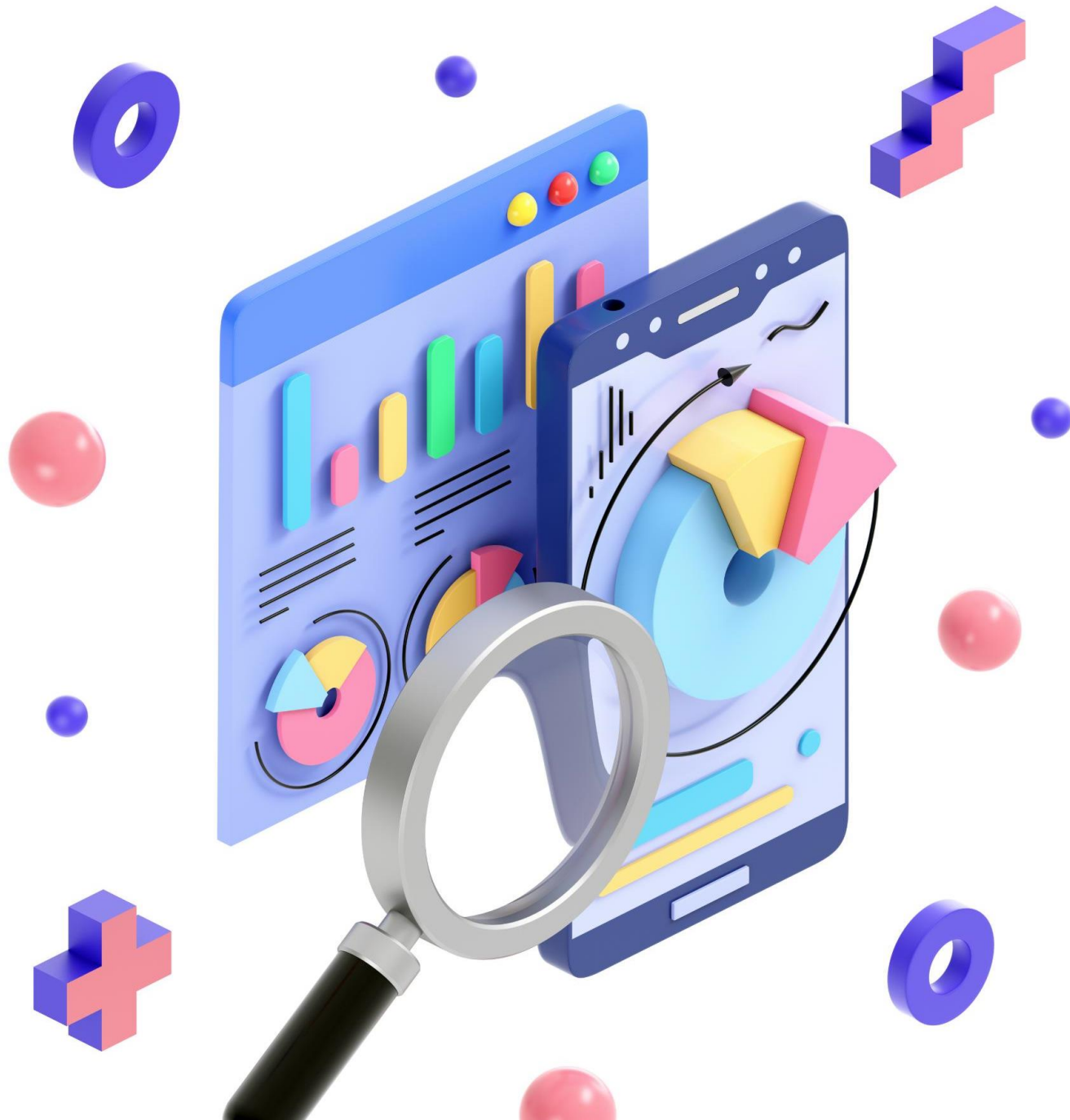
- Focus: Crafting precise instructions
- Challenge: High dependency on exact phrasing

Vibe Coding

- Focus: Shaping tone, style, and “feel” of responses
- Benefit: More creative, less rigid than prompts

Context Engineering

- Focus: Designing the environment around the AI
- Tools: Structured context, memory, workflows



CONTEXT ENGINEERING

Choice of Models and its settings

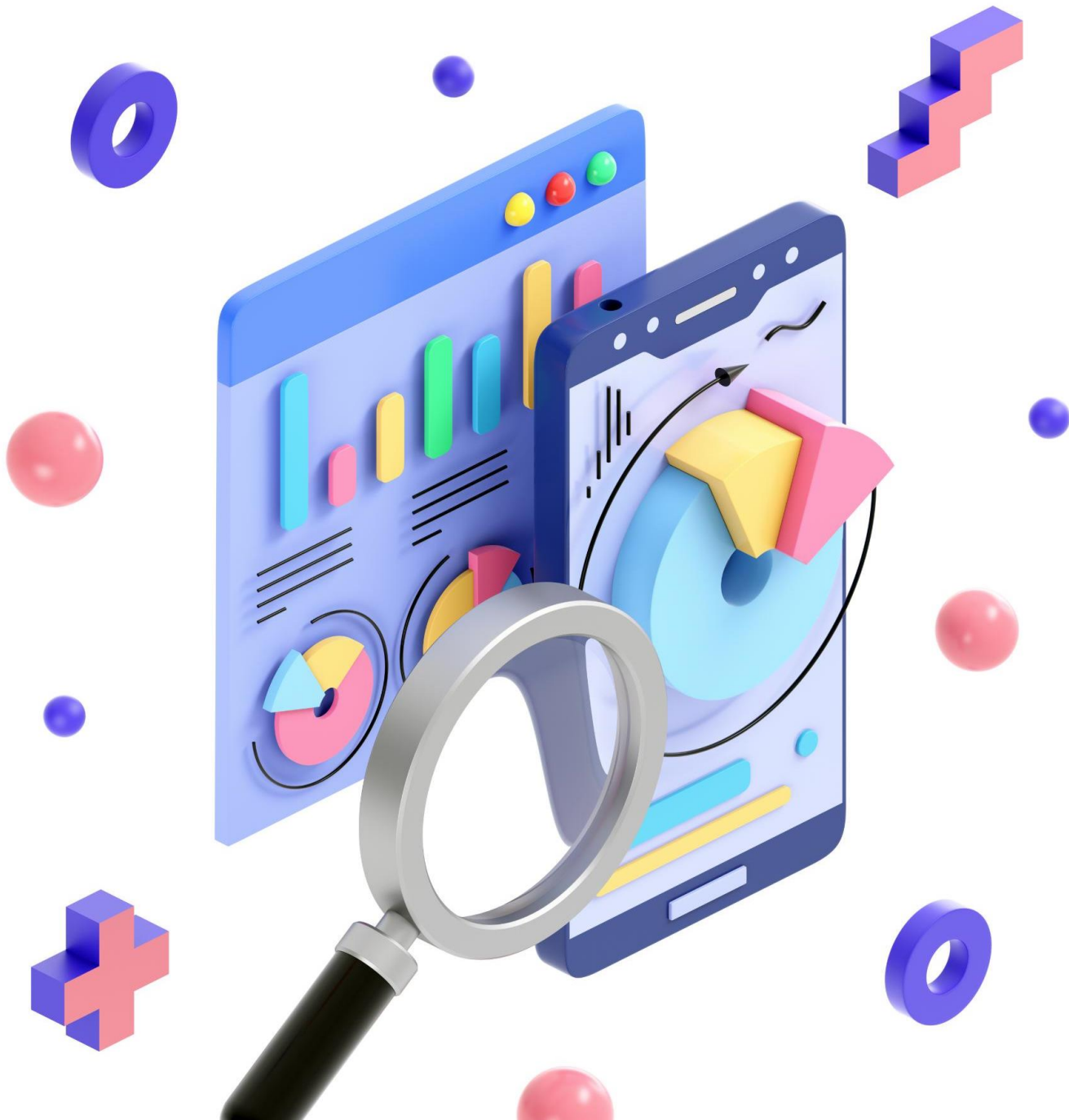
Need clear understanding of how each model behave. Is it proactive or good at following prompts?

Environment setup

The environment has a big effect on the performance. The same model with the same prompt can create wildly different outputs in different environments.

Access to Tools (MCP)

The ability to access tools that could provide additional context and real-time, up-to-date information can improve the Agent response quality.



CONTEXT ENGINEERING

Provided samples

This is how you can teach the Agent of your coding style or unfamiliar patterns

Instructions

The documents will serve as a documentation of all decisions you made about how to develop.

Prompt

This is the last step, the actual prompt. It needs to be clear and concise. Focus on one step at a time especially when you are working on something that is unfamiliar for the AI.

“NON DETERMINISTIC” DEMO TIME





RECOMMENDED CONTEXT FOR POWER BI REPORT DEVELOPMENT

- **Semantic Model**
- **Report design template**
- **Instruction file**
Contains detailed breakdown of the report and model structure
- **Implementation Plan**
Contain a verified step by step approach of the changes you would like the agent to implement.

CONCLUSION

AI Coding Agent

AI tools integration enhances productivity and fosters innovation in Power BI development workflows.

Context Engineering

Following context engineering best practices ensures smooth adoption of AI tools and maximizes business value.

Follow me for more on Data and AI:

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